

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Subject: Python Programming

Assignment No.: 10

Date: 05/01/2026

Lab Assignment: 10

Aim:

To write a Python program to calculate the sum and average of the first 10 natural numbers.

Objective:

To use a `for` loop to find the sum and average of natural numbers from 1 to 10.

Theory:

Natural numbers are positive integers starting from 1.

A `for` loop is used in Python to repeat a block of code for a fixed number of times.

The sum is calculated by adding each number in the loop, and the average is obtained by dividing the sum by the total count.

Algorithm:

1. Start
2. Initialize sum to 0
3. Use a loop from 1 to 10
4. Add each number to sum
5. Calculate average as $\text{sum} \div 10$
6. Display sum and average
7. Stop

Pseudo Code:

START

sum = 0

FOR i = 1 TO 10

 sum = sum + i

END FOR

average = sum / 10

PRINT sum, average

STOP

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Code:

```
sum=0;
for i in range(1,11):
    sum=sum+i;
print("Sum of 1st 10 natural numbers: ", sum);
print("Average of 1st 10 natural numbers: ", sum/10);
```

Output:

```
Sum of 1st 10 natural numbers:  55
Average of 1st 10 natural numbers:  5.5
```

Conclusion:

The program correctly calculates the sum and average of the first 10 natural numbers using a `for` loop.